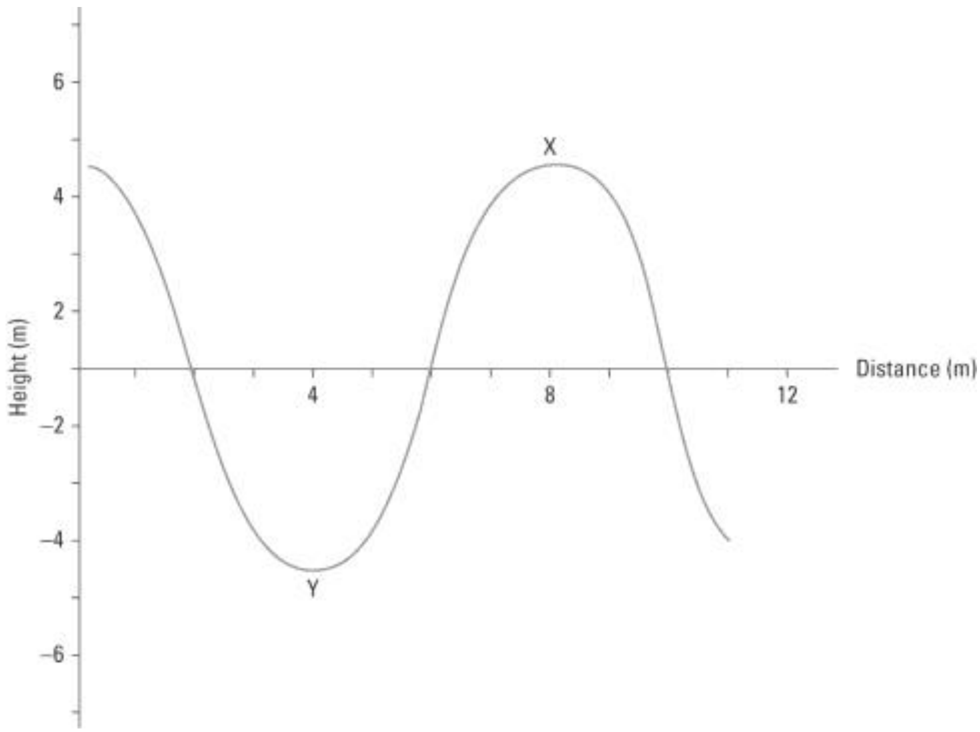
Worksheet 9.6

Waves

Answers

1. Examine the following diagram of a wave.



(a) Identify the type of wave illustrated. ....Transverse..........

(b) Two parts of the wave are labelled X and Y. Name these parts.
X: ....Crest.......... Y: ....Trough..........

(c) Use the scale to determine the wavelength and amplitude of the wave.
Wavelength: ....8 m..........
Amplitude: ....4.5 m (approximate)..........

(d) If the velocity of the wave is 16 m/s, calculate the frequency of the wave.

$v = f\lambda$

$\lambda = 8\text{ m}$

$v = 16\text{ m/s}$

$f = \frac{16}{8} = 2\text{ Hz}$

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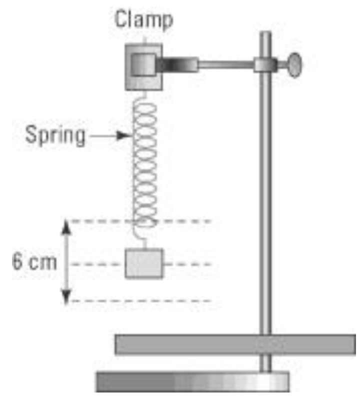
2. Consider the following diagram in which a mass is suspended from the end of a spring that is clamped to a retort stand.



(a) The mass is pulled down and let go. It begins to oscillate up and down. Identify the type of wave motion in the spring.

Compression or longitudinal wave

(b) Determine the amplitude of the vibration.

$\frac{6}{2} = 3$ cm

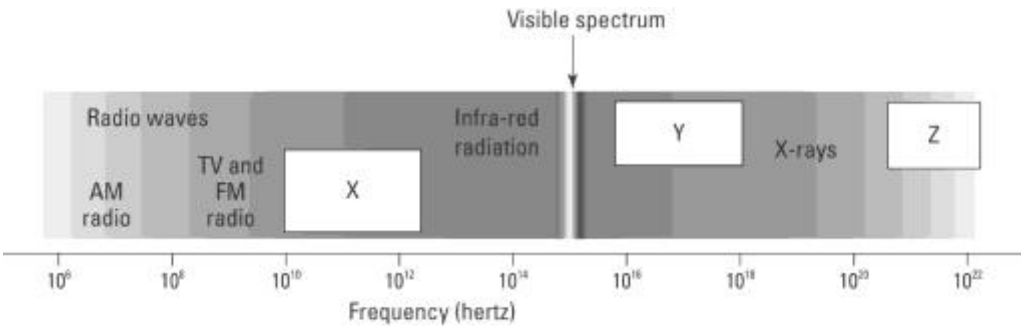
(c) 20 oscillations are made in 5 seconds. Determine the frequency of the oscillation.

$\frac{20}{5} = 4$ oscillations/s $= 4$ Hz

3. Electromagnetic waves are transverse waves which have very high frequencies. Light is an electromagnetic wave. Describe the nature of an electromagnetic wave.

Electromagnetic waves can travel through a vacuum. The waves consist of self-propagating electric and magnetic fields.

4. The following diagram shows the regions of the electromagnetic spectrum. Identify the missing regions X, Y and Z.



X: Microwave (and radar)

Y: Ultraviolet

Z: Gamma

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